# Young Scientist Kenya: One Page Abstract

**Names of students:** Protas Mayaka & Graffins Onyangi

**Category:** Chemical, Physical and Mathematical Sciences

**School:** Kisii School

**County:** Kisii

**Project Title:** Chroma-Scan

**Overview:** Chroma-Scan is a portable digital spectrophotometer designed to bridge the "Scientific Equipment Gap" in Kenyan schools. Commercial spectrophotometers cost over KSH 100,000, forcing students to rely on subjective visual titration. Our project uses an Arduino-based system to measure light absorbance and incorporates a **Machine Learning (ML) Regression Layer** to accurately predict molar concentrations, providing a university-level analytical tool for less than KSH 4,000.

**Background information / Hypothesis:** Precision in chemistry practicals is often limited by human error in color perception. The Beer-Lambert Law states that absorbance is proportional to concentration, but low-cost sensors (LDRs) often introduce non-linear noise. We hypothesize that by integrating an **AI-driven calibration engine**, we can compensate for ambient light interference and sensor drift, achieving a concentration prediction accuracy of over 95%, thereby aligning with the national goal of **Shaping Kenya's AI Future** through low-cost STEM innovation.

**Approach:** 1. **Hardware Construction:** We built a light-isolated chamber using an RGB LED as a quasi-monochromatic source and a high-precision LDR as the transducer.

2. **AI Integration:** We developed a "Neural Calibration" algorithm. Instead of a simple linear formula, the system was trained on a dataset of standard Copper (II) Sulfate ( $CuSO_4$ ) and Potassium Permanganate ( $KMnO_4$ ) solutions to recognize and "filter out" electronic noise.

3. **Software:** The Arduino Nano processes the analog voltage signals, which are then analyzed by the AI layer to display real-time molarity on a 128x32 OLED screen.

4. **Testing:** We compared the AI-enhanced readings against manual visual colorimetry and a standard calibration curve to verify the reduction in "Mean Absolute Error" (MAE).

**Results / Conclusions:** The Chroma-Scan achieved a linear calibration curve with an

$R^2 > 0.985$. The AI integration reduced the error margin caused by ambient light by **15%** compared to traditional digital sensors without ML processing. We concluded that AI-enhanced hardware can democratize high-end chemical analysis in rural schools. By moving from simple sensors to **Predictive STEM tools**, Chroma-Scan proves that Kenyan students can lead the AI revolution using affordable, locally available components.